

COMPUTING THE MORDELL-WEIL GROUP (COMPLETE 2-DESCENT)

E/K elliptic curve, with $E[m] \subseteq E(K)$ for some $m \geq 2$.

• **WEIL PAIRING:** $e_m : E[m] \times E[m] \longrightarrow \mu_m$

• **KUMMER PAIRING:** $k : E(K) \times \text{Gal}(\bar{K}/K) \longrightarrow E[m]$
 $(P, \sigma) \longmapsto k(P, \sigma) = Q^\sigma - Q \quad \text{w/ } [m]Q = P.$

$\longmapsto$ induces: $\delta_E : E(K) /_m E(K) \hookrightarrow \text{Hom}(\text{Gal}(\bar{K}/K), E[m])$

$P \longmapsto \delta_E(P) : G_{\bar{K}} \longrightarrow E[m]$
 $\sigma \mapsto \delta_E(P)(\sigma) = k(P, \sigma)$

• **HILBERT'S THM 90**

$\longmapsto$ induces:
 $(\mu_m \subseteq K)$

$\delta_K : K^\times / (K^\times)^m \xrightarrow{\cong} \text{Hom}(\text{Gal}(\bar{K}/K), \mu_m)$

$b \longmapsto \delta_K(b) : G_{\bar{K}} \rightarrow \mu_m \quad \text{s.t.} \quad \delta_K(b)(\sigma) = \frac{\sigma(\beta)}{\beta} \quad \text{for } \beta^m = b.$

Therefore: $e_m(\overset{K(P, \cdot)}{\underset{\text{Hom}(\overline{G}_K, \mu_m)}{\delta_E(P)}(\cdot)}, T) = \delta_K(b)(\cdot)$
for some $b = b(P, T)$.

Thm. (a) There is a bilinear pairing $b: E(K)/_m E(K) \times E[m] \rightarrow K^\times / (K^\times)^m$
s.t. $b(P, T)$ is the unique class s.t. $e_m(\delta_E(P)(\sigma), T) = \delta_K(b(P, T))(\sigma)$
for all $\sigma \in \text{Gal}(\overline{K}/K)$.

(b) The pairing is non-deg on the left. Thus: $E(K)/_m E(K) \hookrightarrow \text{Hom}(E[m], K^\times / (K^\times)^m)$
 $P \longmapsto b(P, \cdot)$

(c) Let $S = M_K^\infty \cup \{\text{bad primes of } E/K\} \cup \{\text{primes } |m|\} \subseteq M_K$.

Then the image of hom's in pairing b lies in $K(S, m) = \{b \in K^\times / (K^\times)^m : \text{ord}_v(b) \equiv 0 \pmod m \forall v \notin S\}$
(we proved that $K(S, m)$ is finite when proving M-W)

(d) The pairing may be computed: $\text{div}(f_T) = m(T) - m(O)$
 $T \in E[m]$, choose $f_T, g_T \in K(E)$ w/ $f_T \circ [m] = g_T^m$
then $P \neq T$, $b(P, T) \equiv f_T(P) \pmod{(K^\times)^m}$.

Q: How does this help?

$$b(P, T) \equiv f_T(P) \bmod (K^\times)^m \text{ and } b(P, T) \in K(S, m), E[m] = \langle T_1, T_2 \rangle$$

For each $(b_1, b_2) \in K(S, m)^2$ try to solve:

$$\begin{cases} P \in y^2 = f(x) \\ b_1 z_1^m = f_{T_1}(P) \\ b_2 z_2^m = f_{T_2}(P) \end{cases} \Rightarrow \begin{pmatrix} C \\ \text{"} \\ C_{b_1, b_2} \end{pmatrix} \begin{cases} y^2 = x^3 + Ax + B \\ b_1 z_1^m = f_{T_1}(x, y) \\ b_2 z_2^m = f_{T_2}(x, y) \end{cases}$$

$$\text{with } (x, y, z_1, z_2) \in K \times K \times K^\times \times K^\times$$

"homogeneous space"

Pf. (a) $b : E(k)/_m E(k) \times E[m] \rightarrow k^x / (k^x)^m$

Existence + Well-defined follows from Thm 90, i.e., $k^x / (k^x)^m \cong \text{Hom}(G_k, \mu_m)$

κ, e_m are bilinear $\Rightarrow b$ is bilinear.

(b) Suppose $b(P, T) = 1 \quad \forall T \in E[m]$

i.e., $\forall T \in E[m], \forall \sigma \in G_k, e_m(\kappa(P, \sigma), T) = 1$

e_m is non-deg $\Rightarrow \kappa(P, \sigma) = 0 \quad \forall \sigma \Rightarrow P \in \underset{\substack{\text{left ker} \\ \text{of } \kappa}}{mE(k)} \Rightarrow P \equiv 0 \text{ in } E/m.$

(c) $K(S, m) = \{ b \in k^x / (k^x)^m : \alpha Q_\nu(b) \equiv 0 \pmod m, \forall \nu \notin S \}, S = M_k^\infty \cup \{ \text{bad } \nu \mid \nu \mid m \}$

Let $\beta = b(p, T)^{1/m} \Rightarrow K(\beta) \subseteq L = K([m]^{-1} E(k))$

Why? $\left. \begin{aligned} \text{If } \sigma \in G_{\bar{k}/L} \text{ then } \kappa(P, \sigma) = Q^\sigma - Q = 0 \\ \Rightarrow e_m(\underset{\substack{\text{"} \\ \kappa(b(P, T))(\sigma)}}{\delta_E(P)(\sigma)}, T) = e_m(0, T) = 1 \end{aligned} \right\} \begin{aligned} b(P, T) = 1 &= \frac{\sigma(\beta)}{\beta} \\ \Rightarrow \sigma(\beta) &= \beta \Rightarrow \beta \in L. \\ &\forall \sigma \in G_{\bar{k}/L} \end{aligned}$

We just proved $\beta = b(P, T)^{1/m} \Rightarrow k(\beta) \subseteq L = k([m]^{-1}E(k))$

We proved: L/k is unramified outside S

$\Rightarrow$ If $v \in M_k^0$, $v(m)=0$, then $k(\beta)/k$ unram. $\Leftrightarrow \text{ord}_v(b) \equiv 0 \pmod m$

$\Rightarrow b(P, T) \in K(S, m)$. $\square$

(d) $b(P, T) \equiv f_T(P) \pmod{(k^x)^m}$ Let $Q \in E$, $P = [m]Q$, $\beta \in \bar{k}^x$ s.t. $b(P, T) = \beta^m$

$$\forall \sigma \in G_k \quad e_m(\sigma_E(P)(\sigma), T) = f_k(b(P, T))(\sigma)$$

$$e_m(\overset{''}{Q^\sigma - Q}, T) = \frac{\sigma(\beta)}{\beta}$$

$$\frac{g_T(X + \overset{''}{Q^\sigma - Q})}{g_T(X)} = \frac{\sigma(\beta)}{\beta} \xrightarrow{X=Q} \frac{g_T(Q)^\sigma}{g_T(Q)} = \frac{\beta^\sigma}{\beta}$$

δ_k isom: $g_T(Q) \equiv \beta \pmod{(k^x)^m} \Rightarrow g_T(Q)^m \equiv \beta^m \pmod{(k^x)^m}$ (recall: $f_T \circ [m] = g_T^m$)

So $\boxed{f_T(P) = f_T([m]Q) = g_T(Q)^m \equiv \beta^m \equiv b(P, T) \pmod{(k^x)^m}}$ $\square$

COMPLETE 2-DESCENT

$m=2$, $E[z] \subseteq E(K)$, $y^2 = f(x) = (x-e_1)(x-e_2)(x-e_3)$, $e_i \in K$.
 $e_i \neq e_j$

$T_i = (e_i, 0)$ non-trivial 2-tors. pts.

Put $f_{T_i}(x, y) = x - e_i$ then $\text{div}(f_{T_i}) = 2(T_i) - 2(O)$.

and $f_{T_i} \circ [2] = X([z]) - e_i = \frac{(x^2 - 2e_i x - 2e_i^2 + 2(e_1 + e_2 + e_3)e_i - (e_1 e_2 + e_1 e_3 + e_2 e_3))^2}{(2y)^2}$
 $= g_{T_i}^2$

Let $(b_1, b_2) \in K(S, 2) \times K(S, 2)$.

Q Is there $P = (x, y) \in E(K)/2E(K)$ with $b(P, T_1) = b_1$, $b(P, T_2) = b_2$?

$\Leftrightarrow (x, y, z_1, z_2) \in K \times K \times K^\times \times K^\times$

$$\begin{cases} y^2 = (x-e_1)(x-e_2)(x-e_3) \\ b_1 z_1^2 = x - e_1 \\ b_2 z_2^2 = x - e_2 \end{cases}$$

Put z_3 s.t. $y = b_1 b_2 z_1 z_2 z_3$

$$\Rightarrow (b_1 b_2 z_1 z_2 z_3)^2 = b_1 z_1^2 b_2 z_2^2 (x - e_3)$$

$$\Rightarrow b_1 b_2 z_3^2 = x - e_3$$

Cancel
out

$$\Leftrightarrow \begin{cases} b_1 z_1^2 = x - e_1 \\ b_2 z_2^2 = x - e_2 \\ b_1 b_2 z_3^2 = x - e_3 \end{cases} \rightarrow \begin{cases} b_1 z_1^2 - b_2 z_2^2 = e_2 - e_1 \\ b_1 z_1^2 - b_1 b_2 z_3^2 = e_3 - e_1 \end{cases} \quad (\mathcal{C}_{b_1, b_2})$$

Note If we find $(z_1, z_2, z_3) \Rightarrow P = (x, y) = (b_1 z_1^2 + e_1, b_1 b_2 z_1 z_2 z_3) \in E(k)$

• If $P = T$: use linearity:

$$b(T_1, T_1) = \frac{b(T_1, T_1 + T_2)}{b(T_1, T_2)} = \frac{b_1(T_1, T_3)}{b(T_1, T_2)} = \frac{e_1 - e_3}{e_1 - e_2}$$

$$b(T_2, T_2) = \frac{e_2 - e_3}{e_2 - e_1}$$

Thm (Complete 2-descent)

$$E/K : y^2 = (x - e_1)(x - e_2)(x - e_3) \quad e_i \in K, e_i \neq e_j.$$

Let $S = M_K^\infty \cup \{ \text{bad places of } E/K \} \cup \{ \text{places dividing } 2 \}$

Let $K(S, 2) = \{ b \in K^\times / (K^\times)^2 : \text{ord}_v(b) \equiv 0 \pmod{2} \quad \forall v \notin S \}$

Then, there is an injective homomorphism: $E(K)/2E(K) \rightarrow K(S, 2) \times K(S, 2)$

defined by

$$P = (x, y) \mapsto \begin{cases} (x - e_1, x - e_2) & \text{if } x \neq e_1, e_2 \\ \left(\frac{e_1 - e_3}{e_1 - e_2}, e_1 - e_2 \right) & \text{if } x = e_1 \\ \left(e_2 - e_1, \frac{e_2 - e_3}{e_2 - e_1} \right) & \text{if } x = e_2 \\ (1, 1) & \text{if } P = \mathcal{O}. \end{cases}$$

Moreover:

If $(b_1, b_2) \in K(S, 2)^2$ is the image of $P \neq \mathcal{O}, T_1, T_2$ then

$$P = (x, y) \iff \begin{cases} b_1 z_1^2 - b_2 z_2^2 = e_2 - e_1 \\ b_1 z_1^2 - b_1 b_2 z_3^2 = e_3 - e_1 \end{cases} \quad \text{for } (z_1, z_2, z_3) \in K^\times \times K^\times \times K$$

and $P = (b_1 z_1^2 + e_1, b_1 b_2 z_1 z_2 z_3)$.

EXAMPLE (Find $E(\mathbb{Q})$)

$$\text{Let } E/\mathbb{Q} : y^2 = x^3 - 12x^2 + 20x = x(x-2)(x-10)$$

$$\Delta_E = 409600 = 2^{14} \cdot 5^2, \quad C_4 = 2^6 \cdot 3 \cdot 7.$$

There is bad reduction at 2 (additive), 5 (multiplicative).

Torsion: $E[2] \subseteq E(\mathbb{Q})$, more? Recall: $E(\mathbb{Q})[m] \hookrightarrow \tilde{E}(\mathbb{F}_p)$ P is good
PT_m

$$p=3, \quad \# \tilde{E}(\mathbb{F}_3) = 4$$

(good) no prime-to-3 torsion
other than $E[2]$

$$p=11, \quad \# \tilde{E}(\mathbb{F}_{11}) = 16 \Rightarrow E(\mathbb{Q})_{\text{tors}} = E(\mathbb{Q})[2].$$

(good) no 3-torsion.

$$S = \{ \infty, 2, 5 \}, \quad \mathcal{O}(S, 2) = \left\{ b \in \frac{\mathbb{Q}^\times}{(\mathbb{Q}^\times)^2} : \text{ord}_p(b) \equiv 0 \pmod{2} \forall p \notin S \right\}$$

$$\text{Thus,} \quad = \{ \pm 1, \pm 2, \pm 5, \pm 10 \} \quad (\text{size } 8)$$

$$\frac{E(\mathbb{Q})}{2E(\mathbb{Q})} \cong \underbrace{\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z})^{R-1}}_{\text{size } 2^{R+1}} \hookrightarrow \underbrace{\mathcal{O}(S, 2) \times \mathcal{O}(S, 2)}_{\text{size } 8^2 = 2^6} \left. \vphantom{\frac{E(\mathbb{Q})}{2E(\mathbb{Q})}} \right\} \begin{array}{l} R+2 \leq 6 \\ \rightarrow R \leq 4 \end{array}$$

